

TECHNICAL WHITE PAPER v2.0

Methodology and Stochastic Foundations of the LTSA Suite

1. Stochastic Simulation Model

The LTSA business and safety models utilize a multi-parameter Monte Carlo simulation with 1,000+ iterations. We assume a power-law distribution for market adoption and a normal distribution for regulatory approval timelines.

2. PRISMA Literature Synthesis

The v2 literature review utilizes a Grand Synthesis Engine (GSE) pipeline that screens 5,432 unique records using a zero-shot classifier for p-value fidelity and methodological integrity.

3. Genomic Risk Equations

Risk estimation for AAV germline integration is modeled as: $R = (V / B) * (D^k)$, where V is vector titer, B is biological barrier resistance, D is dose, and k is the architecture-specific integration constant.

4. PQC-SCS Security Provenance

All patient telemetry is encrypted via NIST-standard Kyber-1024 and signed via Dilithium-5. The Sovereign Cryptographic Simulation (SCS) ensures forensic data integrity across the global safety mesh.